

// SPATIAL-TEMPORAL ENERGY MODELLING FRAMEWORK

Energy Model for *Renewable Energy* Communities

A full-stack spatial-temporal pipeline that maps building energy demand, renewable generation, and hourly energy balance across Southern Italy — from individual buildings to provincial scale.

Puglia

Calabria

Sicilia

What this model does

This repository implements an end-to-end energy modelling pipeline for assessing Renewable Energy Communities (RECs) in three Southern Italian regions. It integrates building-level spatial data, national census statistics, ARERA domestic consumption records, and renewable plant registries to produce hourly energy balance indicators at the primary substation (cabin) and provincial scale.

Every stage is region-independent: the same SQL schema and Python algorithms run identically for Puglia, Calabria, and Sicilia. Switching regions requires only a schema name substitution.

3

Regions modelled

Puglia · Calabria · Sicilia

5

Pipeline modules

Profile · ARERA · Demand · Supply · Balance

8,760

Hours per year

Hourly resolution · 2024 calendar

Five-stage sequential workflow

Each region runs through the same five stages in order. The ARERA Aggregator and Profile Builder feed data into the Demand Module, which combines with the Supply Module to produce the Energy Balance and KPIs visualised in the dashboard.



Five modules, one cohesive system

Each module is a self-contained processing unit. Together they transform raw geographic, census, and energy data into cabin-level hourly KPIs.



ARERA Aggregator

Python · Shared across all regions

Processes raw monthly electricity withdrawal data for domestic clients from Italy's energy regulator (ARERA). Generates provincial annual kWh averages for all 22 provinces across the study regions.

- | | |
|----|---|
| 01 | ISTAT code mapping for 22 provinces |
| 02 | Batch Excel ingestion per province |
| 03 | Monthly kWh extraction (12 cols) |
| 04 | Annual integral computation |
| 05 | Export consolidated XLSX for PostgreSQL |



Profile Builder

Python · Puglia · Calabria · Sicilia

Ingests raw regional load profiles, normalises hourly coefficients against the annual integral accounting for 2024 FR/FS day calendar, and reshapes into long-format for PostgreSQL injection.

- 01 Calendar config: FR/FS days per month
- 02 Batch Excel → CSV conversion
- 03 Residential kernel: FR/FS separation + normalisation
- 04 Industrial kernel: winter/summer seasonality
- 05 Services/Agriculture kernel: shared diurnal
- 06 Wide → Long reshape (pd.melt)



Demand Module

PostgreSQL + PostGIS · Puglia · Calabria · Sicilia

Maps buildings spatially to census sections, provinces, and primary cabins. Estimates morphological parameters and disaggregates annual energy consumption to hourly 24-element arrays per building.

- 01 GIST spatial indexing + memory config
- 02 Point-in-Polygon join to provinces/cabins/census
- 03 Nearest-neighbour rescue for boundary buildings
- 04 Construction year + floor count estimation
- 05 Volumetric split: res / ser / ind / agr
- 06 Annual kWh → 24h array disaggregation



Supply Module

PostgreSQL + PostGIS · Puglia · Calabria · Sicilia

Spatially maps renewable energy plants (solar, wind, hydro, biomass, geothermal), estimates annual yield via provincial capacity factors, and disaggregates to hourly production arrays.

01 PiP join of plant points to provinces + cabins

02 Topological rescue for orphaned assets

03 Annual yield: potenza × utilisation hours

04 Source classification (solar/wind/hydro/biomass)

05 Annual yield → 24h production arrays



Energy Balance

PostgreSQL · Puglia · Calabria · Sicilia

Aggregates building consumption and plant production to cabin level, computes hourly balance (SC, UD, OP), and derives annual/monthly KPIs with environmental and economic impact metrics.

01 2024 calendar: FR/FS day count table

02 UNNEST demand arrays → cabin hourly long format

03 UNNEST supply arrays → cabin hourly long format

04 $SC = \min(D, S)$ · $UD = \max(D - S, 0)$ · $OP = \max(S - D, 0)$

05 Annual + monthly SCI, SSI at cabin/province

06 OPI, avoided CO₂, economic benefit, env nexus



WebGIS Dashboard

Streamlit · GeoPandas · Plotly Express

Interactive choropleth web application that renders SCI, SSI, and OPI at the primary substation level across Southern Italy. Reads from GeoJSON exported by the Energy Balance module.

01 Load cabins_kpi.geojson → GeoDataFrame

02 Reproject to EPSG:4326 (WGS 84)

03 Coerce KPI columns to numeric

04 Choropleth mapbox (Viridis scale)

05 Interactive KPI selectbox + hover tooltips

Three KPIs, one energy story

Every primary substation area and province gets three complementary indicators that together describe the energy autonomy and efficiency of a potential REC.

SCI

Self-Consumption Index

$$\frac{\sum SC_h}{\sum P_h}$$

$$\sum P_h$$

Share of local production that is consumed locally. A high SCI means most generated energy never reaches the grid.

SSI

Self-Sufficiency Index

$$\frac{\sum SC_h}{\sum D_h}$$

$$\sum D_h$$

Share of local demand covered by local production. A high SSI means the community is largely energy-independent.

OPI

Over-Production Index

$$\frac{\sum OP_h}{\sum P_h}$$

$$\sum P_h$$

Share of production that exceeds local demand. High OPI signals infrastructure oversizing or export opportunities.

Beyond energy — the environmental nexus

The Energy Balance module computes avoided emissions and resource savings at provincial level, replacing the fossil-heavy national grid mix with local RES. Emission factors are based on the Italian national grid profile.



CO₂ equivalent

256

tons / GWh



NO_x avoided

0.4

tons / GWh



SO₂ avoided

0.15

tons / GWh



PM₁₀ avoided

0.02

tons / GWh



Water savings

2,500

m³ / GWh



SC value

0.22

€ / kWh saved



OP shared value

0.10

€ / kWh shared

Renewable source classification

The Supply Module maps Italian GSE source codes (*ai_fonte*) to five RES types:

ITALIAN SOURCE (AI_FONTE)	MAPPED TYPE	YIELD DRIVER
SOLARE	solar	Provincial solar irradiance (utilisation hours)
EOLICA	wind	Provincial wind speed (utilisation hours)
IDRAULICA	hydroelectricity	Hydrology factor
BIOGAS · BIOMASSE · RIFIUTI	biomass	Capacity factor (near-constant)
GEOTERMICA	geothermal	Base-load factor

What the model is built on

COMPONENT	TECHNOLOGY	ROLE
Spatial database	PostgreSQL 15PostGIS 3	All spatial joins, building tables, hourly arrays
Spatial indexing	GIST R-tree	High-performance Point-in-Polygon queries
Data processing	Python 3.11pandasnumpy	Profile normalisation, ARERA extraction
Dashboard	StreamlitGeoPandasPlotly	Interactive choropleth WebGIS
Coordinate system	EPSG:4326	WGS 84, standard web mapping CRS
Array architecture	real[] arrays	24-element hourly vectors per building/month

Data sources

SOURCE	DESCRIPTION	COVERAGE
ISTAT Census 2011	Section-level building indicators (E8–E20)	National
ISTAT Census 2021	Dwelling counts per section (A8)	National
ISTAT prov_2024	Provincial administrative boundaries	National
ARERA		Provincial

SOURCE	DESCRIPTION	COVERAGE
	Monthly per-capita domestic electricity withdrawal	
ENEA	Sectoral regional consumption (industry, services, agr)	Regional
GSE / MiTE	Renewable plant registry (ai_fonte, potenza)	National
Municipal cadastre	Building classification, use type, height	Regional
Primary cabins 2025	Primary substation service areas (cod_ac)	Regional

Execution order

Run these steps in sequence for each region. The schema name ([sicia](#)) is the only substitution needed to switch regions.

01

Normalise temporal profiles (Profile Builder)

Reads raw ARERA load profile Excel files and outputs a normalised long-format CSV for PostgreSQL.

```
python Profile_Builder_Sicilia/profile_builder.py
```

02

Aggregate ARERA domestic consumption

Extracts provincial monthly kWh from ARERA Excel files and consolidates into a single XLSX for PostgreSQL ingestion.

```
python ARERA_Domestic_Consumption_AGGREGATOR/arera_aggregator.py
```

03

Run the Demand Module (SQL)

Performs spatial mapping, morphological estimation, and hourly consumption disaggregation for all buildings.

```
psql -d your_database -f Demand_Module_Sicilia/demand_module.sql
```

04

Run the Supply Module (SQL)

Spatially maps RES plants, estimates annual yield, and disaggregates to hourly production arrays.

```
psql -d your_database -f Supply_Module_Sicilia/supply_module.sql
```

05

Compute Energy Balance and KPIs (SQL)

Aggregates to cabin level, computes SC/UD/OP per hour, and derives SCI, SSI, OPI with environmental metrics.

```
psql -d your_database -f Energy_Balance_Sicilia/energy_balance.sql
```

06

Launch the WebGIS Dashboard

Serves an interactive choropleth map of SCI, SSI, and OPI at cabin level across all three regions.

```
streamlit run app/app.py
```

Prerequisites

```
# PostgreSQL with PostGIS
```

```
sudo apt install postgresql-15 postgresql-15-postgis-3
```

```
# Python dependencies
```

```
pip install pandas numpy openpyxl geopandas streamlit plotly
```

REC Energy Model · Southern Italy · Reference year 2024 · Spatial CRS EPSG:4326 ·
Temporal resolution 8,760 h/yr

PostgreSQL + PostGIS · Python · Streamlit · Plotly · GeoPandas



Note: The initial draft of this README documentation was generated with the assistance of Claude AI and subsequently reviewed, verified, and edited by the author for accuracy.